Homework 0

Numerical Solutions to Differential Equations

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Contents

1 Problem 1 **2**

2 Problem 2 **4**

3 Problem 3 **6**

4 Problem 4 **8**

1 Problem 1

Problem Statement:
Write a matlab code that uses finite differences to compute the first derivative of

$$f(x) = \exp(\sin(x))$$

on the internal [0,2π]. se second-order centered differences on internal grid points and first-order one-sided differences at endpoints.

1. Show a graph of the exact analytic solution $f'(x)$ and your numerical solution $f'_n(x)$.
2. Show a relative error convergence plot by plotting the relative error on the y-axis and the grid size N on the x-axis. Plot the error using both the $p = \infty, 2$ norms in a log-log scale.
3. We see a curve with a slope 3/2, why?

Solution:

1. A Finite difference approximation of the first derivative of $f(x) = \exp(\sin(x))$ was taking using N = 21. The results are shown in figure 1 below.

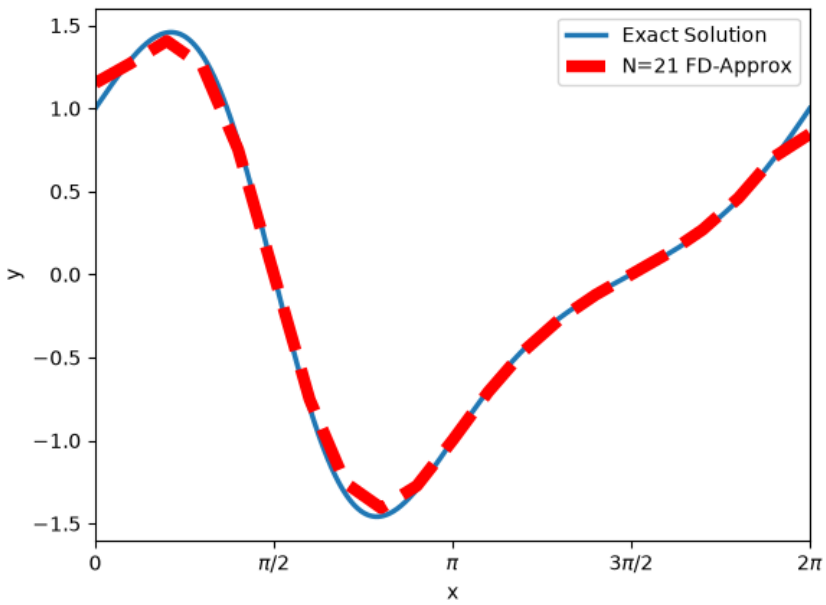


Figure 1: Finite difference approximation (N=21) of first derivative of $f(x) = \exp(\sin x))$. The exact solution is shown in blue and the numerical solution is shown in red.

2. To demonstrate the convergence of the finite difference model, a convergence plot is provided in figure2. Notice that the curve for the L_2 norm (shown in blue) shows a slope of -3/2, while the L_∞ norm yields a curve with a slope of -1. The slope of -1 for L^∞ is expected, as the forward differences used for the endpoints of the numerical approximation have an error that is $O(h)$, where h represents the distance between nodes. However, the slope of -3/2 for the L^2 error is unexpected. This is discussed in part (c).

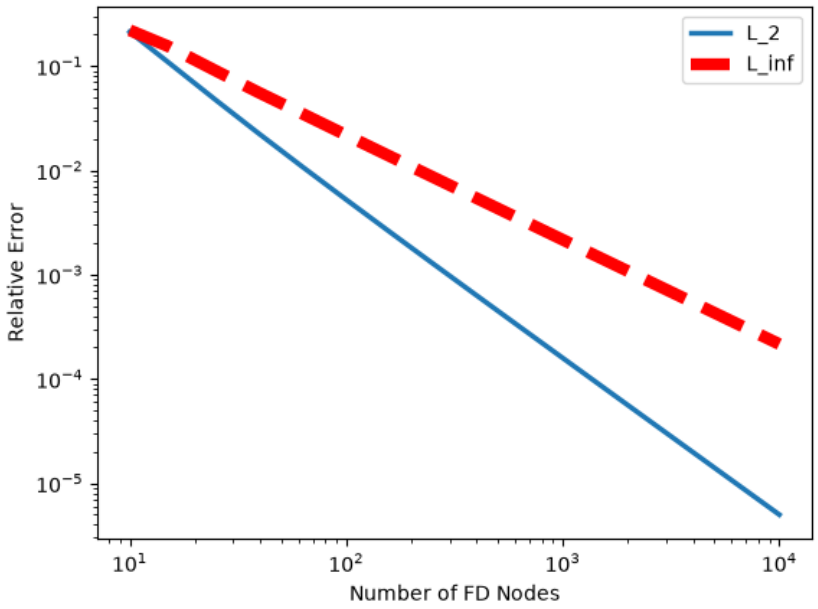


Figure 2: Convergence plot with L^2 and L^∞ norms of absolute error for finite difference approximation shown in figure 1.

3. The absolute error for the case where the derivative at the end points is approximated by first order finite differences can be rewritten using asymptotics as follows:

$$\frac{\|f(x_i) - f_n(x_i))\|_2}{\|f(x_i)\|_2} = \frac{\sqrt{(\mathcal{O}(h) + \mathcal{O}(n)\mathcal{O}(h^2))^2}}{\sqrt{\mathcal{O}(n)}} = \frac{\mathcal{O}(h)}{\sqrt{\mathcal{O}(n)}}$$

Now note that $\mathcal{O}(n)$ grows like $\mathcal{O}(h^{-1})$, so error can be further simplified

$$\frac{\mathcal{O}(h)}{\sqrt{\mathcal{O}(h^{-1})}} = \mathcal{O}(h^{3/2})$$

2 Problem 2

Problem Statement:
Repeat problem 1 but compute the second derivative f"(x). Assume periodicity and wrap around the grid instead of using one-sided finite differences.

Solution:

1. A Finite difference approximation of the first derivative of f (x) = exp(sin(x)) was taking using N = 21. The results are shown in figure 3 below.

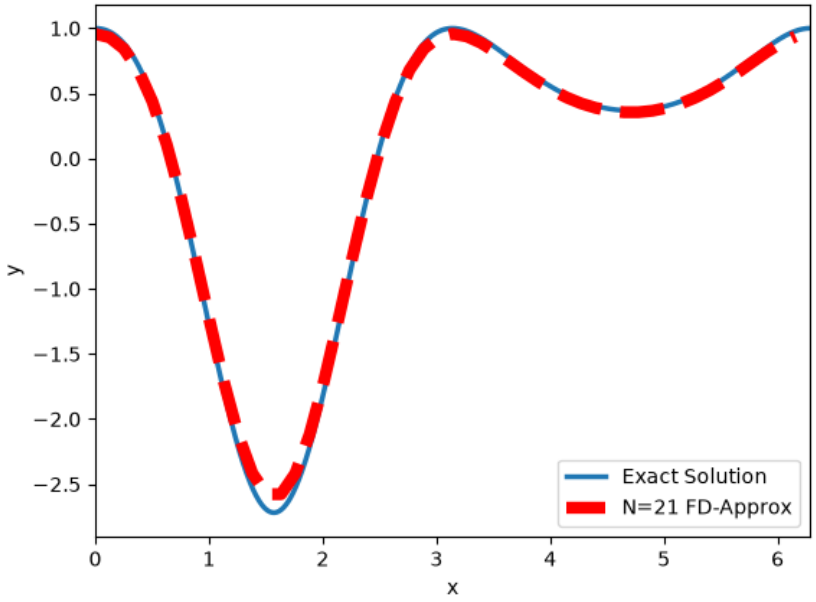


Figure 3: Finite difference approximation (N=20) of second derivative of $f(x) = \exp(\sin x))$. The exact solution is shown in blue and the numerical solution is shown in red.

2. Figure 4 below shows the L^2 error for the finite difference approximation of the second derivative of f. This curve has a slope of -2, which is in line with the second order convergence of the centered difference formula.

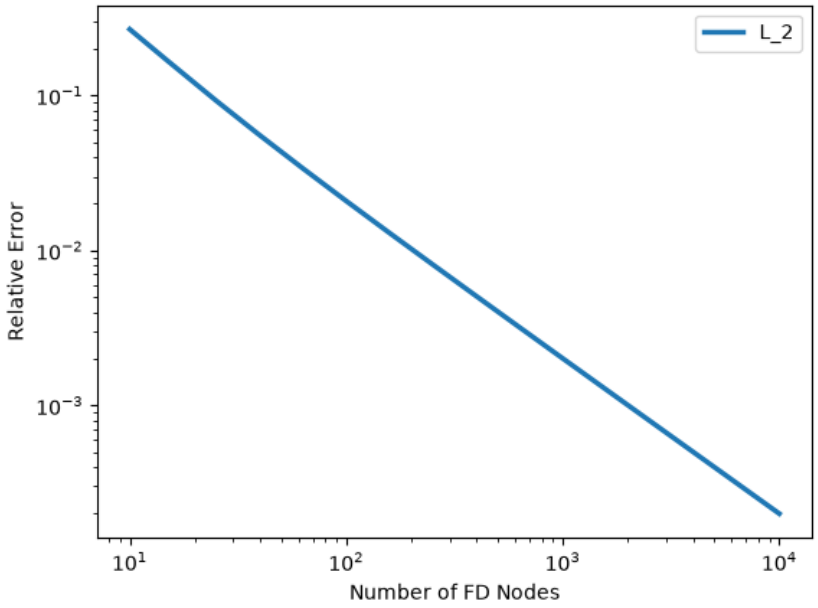


Figure 4: Convergence plot with L^2 norms of absolute error for finite difference approximation shown in figure 3.

3 Problem 3

Problem Statement:
Consider the ODE

$$\frac{d^2u}{dx^2} + sin(x)\frac{du}{dx} + u(x) = f(x) \tag{1}$$

defined on $0 \leq x \leq 5$. Solve the ODE using second-order finite difference methods with the Dirichlet conditions. Use $u(x) = sin(x)$ to test your solution by finding boundary conditions and the necessary forcing function $f(x)$. Show an error plot that proves your method converges with second order.

Solution: Applying finite differences to equation 1 we can rewrite the problem as

$$u_{i+1}\left(\frac{1}{h^2} + \frac{\sin(x_i)}{h}\right) + u_i\left(1 - \frac{1}{h^2}\right) + u_{i-1}\left(\frac{1}{h^2} - \frac{\sin(x_i)}{h}\right) = \sin(x_i)\cos(x_i), \tag{2}$$

where x_i, u_i denote the values of x and u at the ith node of the finite difference mesh. u_0 and u_n are held at the appropriate boundary values. Solving the system defined by equation 1 for n = 50 gives the result shown in 5.

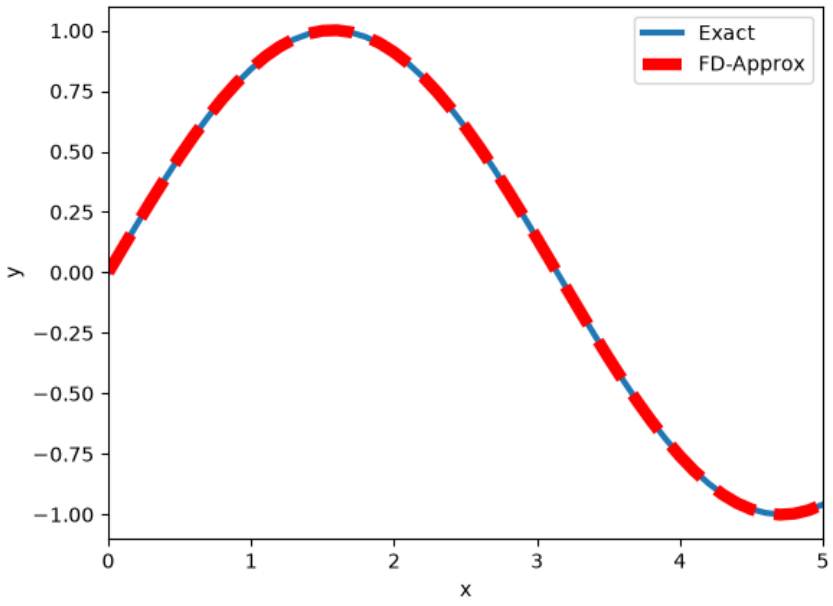


Figure 5: Finite difference approximation (n=50) of solution to equation 1 The exact solution is shown in blue and the numerical solution is shown in red.

Plotting the error, it can once again be seen that the finite difference approximation has second order convergence.

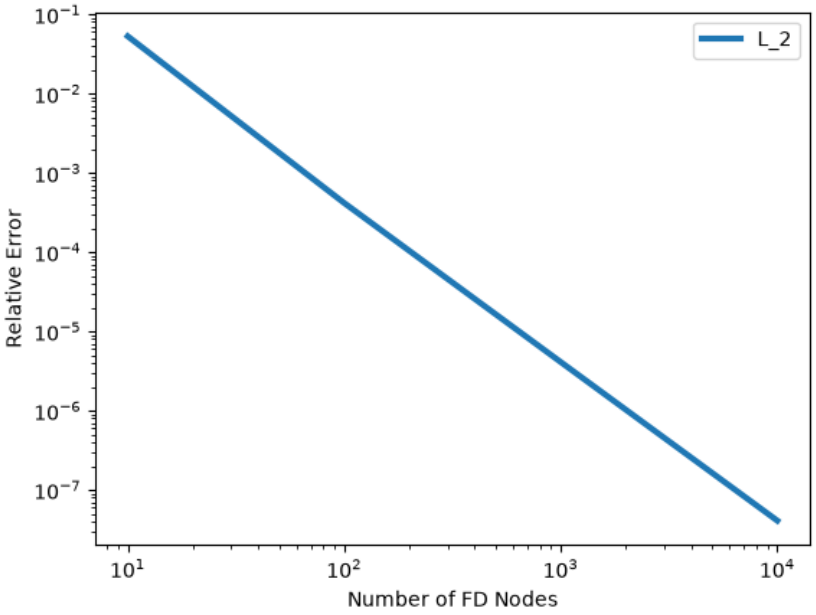


Figure 6: Convergence plot with L^2 norms of absolute error for finite difference approximation in equation 2.

4 Problem 4

Problem Statement:
Use finite difference methods to solve the Poisson equation $\nabla^2 u = f(x, y)$ in the domain $x \in [0, 5],\ y \in [0, 5]$. Use the test solution $u(x, y) = sin(x)cos(y)$

1. Apply Dirichlet conditions on all boundaries so that $u(x, y)$ satisfies the PDE and the boundary conditions, and show a plot of your solution and a second order convergence plot (similar to the first three problems).

2. Modify your code to apply Neumann conditions on all boundaries. This won't work. Why not? Both in terms of the linear algebra and in terms of the actual PDE we want to solve. Suggest a solution to the issue.

Solution:

1. Figure 7 below depicts the exact and numerical solutions to the Poisson equation with $f(x) = -2\sin(x)\cos(x)$ for n = 1600. Using the eyeball norm, the solutions are identical.

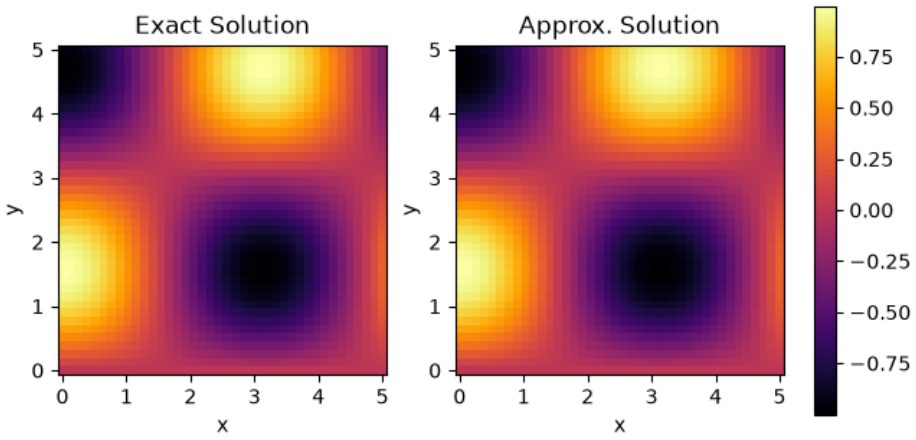


Figure 7: The above figure depicts the exact and numerical solutions to the Poisson equation.

Since the error in the numerical solution is not easily visible by inspection, figure 8 provides an error plot.

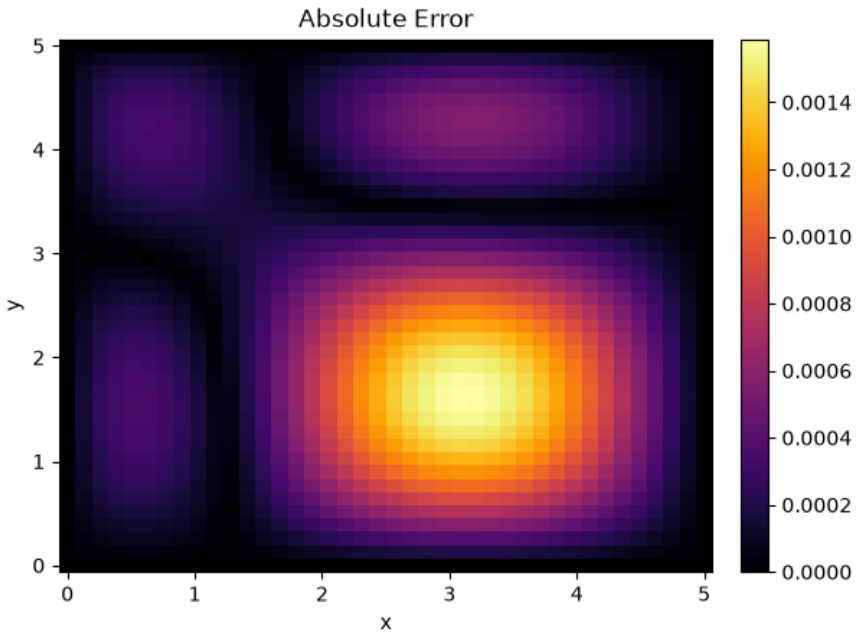


Figure 8: The figure above depicts the error in the FD approximation to the solution of the Poisson Equation.

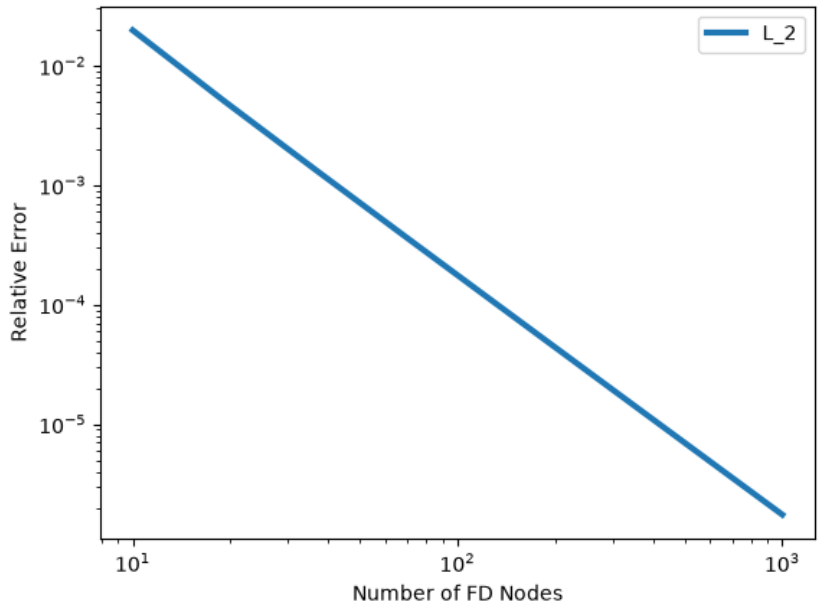


Figure 9: Convergence plot with L^2 norms of absolute error for finite difference approximation of the Poisson equation.

Running the numerical simulation for denser and denser meshes, it is found that the finite difference model once again demonstrates second order convergence, as shown in figure 9

2. Attempting to adapt the FD solver to handle the same problem as above but with Neumann boundary conditions results in the solution shown in figure 10.

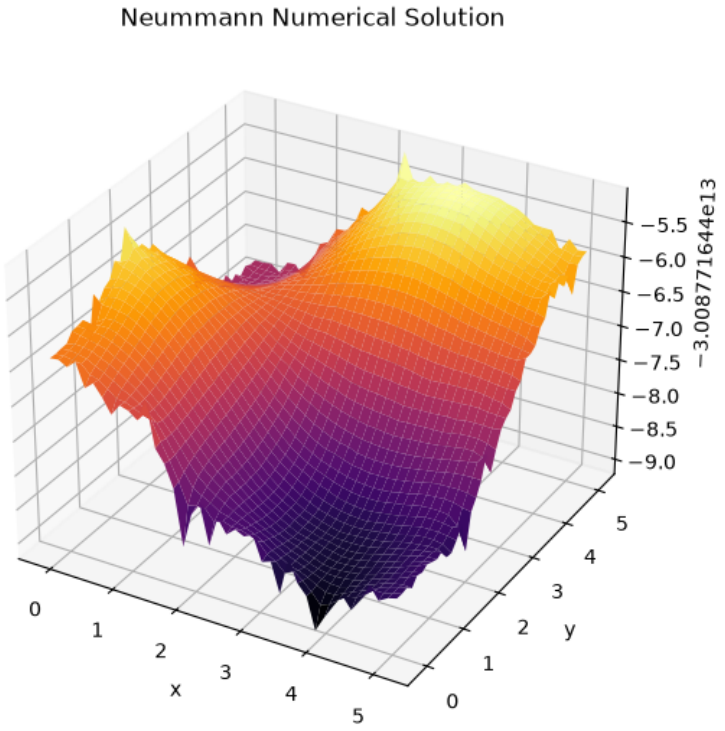


Figure 10: Convergence plot with L^2 norms of absolute error for finite difference approximation in equation 2.

As the figure above makes clear, the finite difference solver diverges for these boundary conditions. This is because linear system used to approximate the laplacian operator becomes singular when neumann boundaries are applied. Numpy estimated the condition number of the corresponding A matrix for a 40 by 40 grid to be 1.17e17. This occurs because there are infinite solutions to Poisson's equations when the only specified boundary conditions are Neumann. To address these issues there are two paths forward:

(a) Gather more data regarding the system being analyzed to impose additional conditions on the solution to narrow the solution space to one answer (This would be the best path forward if you are trying to model a system which you can gather data about).

(b) Impose an additional constraint on the loss function used to solve the poisson equation to select an "optimal" solution from the solution space (This would be the best fit if you are trying to solve a design problem where multiple solutions are valid).